Beginning Matlab/Numerical Computation Software

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Goals.

- Be able to perform the basic Matlab operations.
- Be able to use 'HELP' whenever in need.

Matlab uses three primary windows (We only introduce the first two windows today!):

- Command window. Used to enter commands and data.
- Graphics window. Used to display plots and graphs.
- Edit window. Used to create and edit M-files.

1 Predefined variables and the number of significant digits

There are several predefined variables (see Table 1), for example, `pi`.

```
>> pi

ans =

    3.1416
```

It displays only 5 significant digits. If you desire additional precision, enter the following:

```
>> format long
```

Now when pi is entered the result is displayed to 16 significant figures:

```
>> pi

ans =

   3.141592653589793
```

To return to the 5 decimal version, type

```
>> format short
```

Below we can find how a real number store in Matlab with 5 decimal digits. We type `1.23456` and enter it:

```
>> 1.23456

ans =

    1.2346
```

We type `1.23453` and enter it:

```
>> 1.23453

ans =

    1.2345
```

It shows: The real number is stored or presented by **round-off** in MATLAB with 5 decimal digits.

2 Arithmetic operations

MATLAB can be used as a calculator. Some of the problems indicate the precedence of arithmetic operations.

1. `>> 3*2^4`
2. `>> (3*2)^4       % parentheses have highest priority`
3. `>> 3-2^4`
4. `>> 3^4-3`
5. `>> 8/2^4`
6. `>> 8^4/2`

Precedence of arithmetic operations is: (1) parentheses, (2) exponentiation, (3) multiplication and division, and (4) addition and subtraction. Calculations proceed from left to right on the line and equal precedence is settled in this way.

3 Variable, vector, matrix

Example 3.1. (Variables)

$$\begin{aligned} a &= 1 \\ b &= 2; \\ c &= a + b \end{aligned}$$

Comment: With/without the semicolon: the computer does not/does outputs the results.
Comment: Matlab is case sensitive in defining variables.
Comment: Use 'clear' to clear definition of a variable. Use 'clc' to clear screen.

1. Create vectors

1. `>> x = [3 4 7 11]   % create a row vector (spaces)`
2. `>> x = 3:8            % colon generates list; default stride 1`
3. `>> x = 8:-1:0         % ⟨start⟩ : ⟨stride⟩ : ⟨stop⟩ specifies list`
4. `>> xx = [ 8 7 6 5 4 3 2 1 0];     %  same as last; semicolon suppresses output`
5. `>> xx                   % display contents`
6. `>> x = linspace(0,1,11)  % generate vector automatically`
7. `>> x = 0:0.1:1           % same thing`
8. `>> y = linspace(0,1);    % note semicolon!`
9. `>> length(x)`
10. `>> length(y)`

```
>> size(x)
>> size(y)
>> y(3)                    % access single element
>> y(1:12)                 % access first twelve elements
>> y([3 6 9 12])           % access values specified in a vector!
>> x'                      % transpose
>> 3*[1 2 5]               % factor replicated, multiplies each element
```

Matlab labels the arrays (vectors and matrices) beginning with 1; this will be an important programming detail on more than one occasion.

3.1 Vector and matrix arithmetic

We'll now explore making matrices and doing arithmetic with them. There are some new arithmetic operations that you did not see before in linear algebra, but they will be useful nonetheless.

```
g = [1; 2; 3; 4]
g = [1 2 3 4; 5 6 7 8]
g - 2
2*g-1
g = [1 2 3 4; 5 6 7 8; 9 10 11 12]
h = [1 1 1 1; 2 2 2 2; 3 3 3 3]
g-h
g*h                % What happened?
h*g
g.*h               % NOT the usual matrix multiplication! Elementwise!
g./h               % elementwise division; denominator below slash
h.\g               % Last two same!
g*h'
g'*h
e = [ 1 2 3; 4 5 6; 7 8 0]
f = [9 8 7; 6 5 4; 3 2 1]
e*f
f*e                % Not same!
q = rand(3)        % matrix with elements uniformly on [0,1]
A^2                % raise matrix to a power
A^2 - A*A          % verify with definition
A.^2               % square each element
A.^2 - A*A         % not the same!
```

4 Plotting

Let me show you and example of how to manually plot a function

```
>> f = @(x) x.^2-3*x.*sin(x);  % @(x) says x is the variable;
>> f(0)                        % Evaluate f at 0
>> f(1)                        %
>> f(2)
>> f(0:2)                      % We can do this because of .* and .^.
                               % See what happens if there is no . in .^*.

>> help fplot
>> fplot(f)

>> x = 0:0.01:6*pi;
>> plot(x,f(x))

>> figure                      % plot a new figure
>> g = @(x) (1 + 1./x - log(x) )./(1+x).^2;
>> x = 1:0.01:6;
>> plot(x,g(x))
```

Copy it and run it.

Let us see the 3D plot of the Mobius Strip.

```
% given variables u and v
[u, v] = meshgrid(linspace(0,2*pi,100), linspace(-1,1,50) );
% functions
x = ( 1 + v/2 .*cos(u/2) ).* cos(u);
y = ( 1 + v/2 .*cos(u/2) ).* sin(u);
z = v/2 .* sin( u/2 );
% use function surf to plot a 3D figure
figure
surf(x,y,z);
axis equal;
% add title and labels of axis
title('Mobius Strip')
xlabel('x')
ylabel('y')
zlabel('z')
% view direction
view(3)
```

5 Functions

Example 5.1. (addition and subtraction)

Function name: add sub

Input: a, b, sign

Output: c

**********MATLAB CODE **********

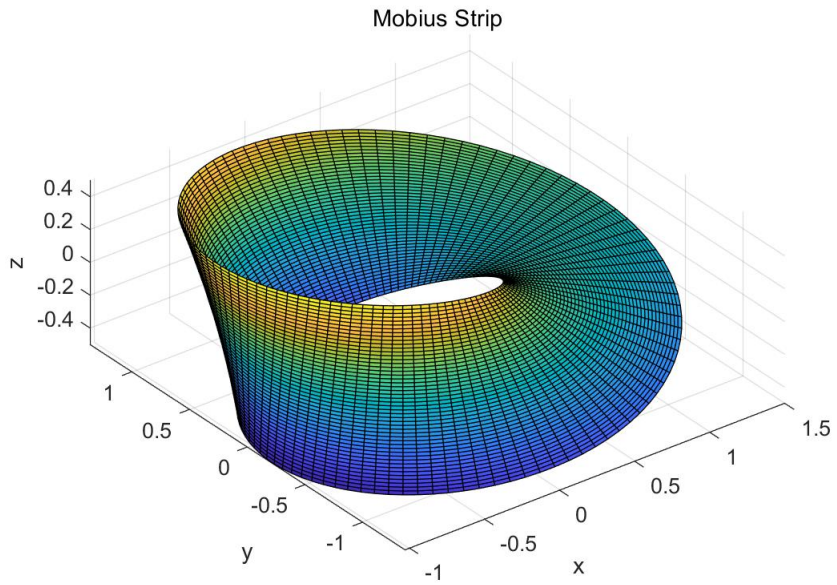
function [c] = add_sub(a, b, sign)

if(sign==1)

c = a+b;

else

Figure 1: The figure of the Mobius Strip.




```
c = a-b;
end
end
*****************************************************************
```

6 Repeating Commands I

Table 1: A list of predefined values and variables in MATLAB.

Expression	Description
pi	The number π up to 15 significant digits.
i, j	The complex number $\sqrt{-1}$.
inf	Represents the mathematical Infinity concept, for example, a result of division by zero.
NaN	Stands for Not-A-Number. Represents the result of a meaningless mathematical function, like 0/0.
clock	Contains the current date and time in the form of a 6-element row vector: year,month,day,hour,minute,second.
date	Contains a string representing today's date.
eps	Stands for epsilon. It represents the smallest number that can be represented by your MATLAB software.
ans	A special variable that MATLAB uses to store the result of MATLAB's command line.

7 Reading Assignments/References:

1. Numerical Computation Software
2. Matlab Tutorial: there are many; e.g., follow the link Mathworks tutorial , or MIT MATLAB Tutorial, or simply google 'Matlab Tutorial' ! Also, here is a link to Notes on Computational Mathematics: Matlab written by Prof. Higdon.

3. More numerical analysis explorations and examples in MATLAB can be found via Professor Cleve Moler's website.